Designing Robots 🛠️

Junior Robotics Badge — Girls Build Bots

Presented by your robotics mentors!

Robot Anatomy 🦾

Every robot has five part groups:

Part	Like your…	Job
Sensors	eyes & ears 👀	detect the world
Controller	brain 🧠	make decisions
Chassis	body 🦴	hold it together
Motors	muscles 💪	create movement
Effectors	hands 🖐️	do the actual job

Anatomy Quiz Game! 🎯

Mentor names a robot — you name a part and its job!

- 🚀 Mars rover → *“Cameras! They’re sensors that see rocks!”*
- 🏠 Robot vacuum → ?
- 🏭 Factory arm → ?
- 🐕 Robot dog → ?

The Engineering Design Process

How real engineers work:

DEFINE the problem → **BRAINSTORM** ideas → **PLAN** your design → **BUILD** it → **TEST** it → **IMPROVE** it → *(and loop back!)*

You'll use ALL these steps today.

Design Sprint: Robots That Help 💗

In teams of 2–3, pick a mission: - 🛒 Help an elderly neighbor carry groceries - 🐈 Find a lost pet - 🌱 Water a garden automatically - 🗑️ Clean up a park

Step 1–2: Define & Plan (25 min)

On your planning sheet: 1. Who are you helping? What's their problem exactly? 2. **Sketch your robot** and label: - Its sensors - Its brain - How it moves - Its tool 3. List your materials

Engineers plan BEFORE they build!

Step 3: Build (30 min) 🔨

Build your model! Rules: - Every labeled part must be represented (a bottle cap can be a camera!) - Everyone on the team builds - It doesn’t have to work — it has to **show your design**

Step 4: Test & Improve (15 min) 🧪

A mentor will visit as your **skeptical customer**:

> *“Hmm… what happens when it rains?” “How does it know where the garden ends?”*

Answer their hard question, then make **ONE improvement** to your design.

Design Review Circle ⭐

Each team: 90-second pitch! 1. The problem 2. The parts 3. The improvement you made

Audience gives **Two Stars and a Wish**: ⭐ ⭐ two things you liked +  one suggestion

You did it! 🎉

Today you: - ✅ Learned all five robot part groups - ✅ Used the full engineering design process - ✅ Built and improved a prototype - ✅ Gave and received engineering feedback

Keep going at home! 🏠

- **Tinkercad** (tinkercad.com) — free 3D design tool
- **NASA Space Place** (spaceplace.nasa.gov) — space robots explained
- Instructables.com — kid-friendly build projects (with a grown-up)
- Next badge: **Showcasing Robots**!